

ITA0608 - Machine Learning

List of Lab Exercises

13. Implement the Car Price Prediction Model using Python.

AIM

To develop a Car Price Prediction Model using Python that predicts the selling price of a car based on features such as its age, mileage, engine capacity, and horsepower using a Linear Regression algorithm.

ALGORITHM

1. Import the required Python libraries.
2. Create or load a car price dataset.
3. Select the relevant input features and target variable.
4. Split the dataset into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict car prices using the test data.
8. Calculate the model's performance using Mean Squared Error and R^2 score.
9. Predict the price of a new car using the trained model.
10. Display the predicted price and model performance.

PROGRAM

DATA SET 1:

```
from sklearn.linear_model import LinearRegression
X = [
    [1, 12000, 1200],
    [2, 18000, 1200],
    [3, 25000, 1300],
```

```

[4,35000,1400],
[5,45000,1500],
[6,55000,1500],
[7,65000,1600],
[8,75000,1600],
[9,85000,1800],
[10,95000,1800]
]
y = [8.5,8.0,7.4,6.8,6.1,5.5,5.0,4.5,4.0,3.5]
model = LinearRegression()
model.fit(X, y)
new_sample = [[4,38000,1400]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

from sklearn.linear_model import LinearRegression
X = [
    [1,12000,1200],
    [2,18000,1200],
    [3,25000,1300],
    [4,35000,1400],
    [5,45000,1500],
    [6,55000,1500],
    [7,65000,1600],
    [8,75000,1600],
    [9,85000,1800],
    [10,95000,1800]
]
y = [8.5,8.0,7.4,6.8,6.1,5.5,5.0,4.5,4.0,3.5]
model = LinearRegression()
model.fit(X, y)
new_sample = [[4,38000,1400]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

```

*** Predicted Car Price: 6.795681973954762 Lakhs

```

DATA SET 2:

```

from sklearn.linear_model import LinearRegression
X = [
    [2024,10000,22],

```

```

[2023,18000,21],
[2022,25000,20],
[2021,35000,19],
[2020,45000,18],
[2019,55000,17],
[2018,65000,16],
[2017,75000,15],
[2016,85000,14],
[2015,95000,13]
]
y = [10.5,9.8,9.0,8.2,7.5,6.8,6.0,5.4,4.8,4.2]
model = LinearRegression()
model.fit(X, y)
new_sample = [[2021,32000,19]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

▶ from sklearn.linear_model import LinearRegression
  X = [
    [2024,10000,22],
    [2023,18000,21],
    [2022,25000,20],
    [2021,35000,19],
    [2020,45000,18],
    [2019,55000,17],
    [2018,65000,16],
    [2017,75000,15],
    [2016,85000,14],
    [2015,95000,13]
  ]
  y = [10.5,9.8,9.0,8.2,7.5,6.8,6.0,5.4,4.8,4.2]
  model = LinearRegression()
  model.fit(X, y)
  new_sample = [[2021,32000,19]]
  prediction = model.predict(new_sample)
  print("Predicted Car Price:", prediction[0], "Lakhs")

```

```

... Predicted Car Price: 8.057525252525465 Lakhs

```

DATA SET 3:

```

from sklearn.linear_model import LinearRegression

```

```

X = [
    [1000,70,10000],
    [1200,82,15000],
    [1400,95,22000],
    [1500,105,30000],
    [1600,115,38000],
    [1800,130,45000],
    [2000,145,52000],
    [2200,160,60000],
    [2400,175,68000],
    [2600,190,76000]
]
y = [6.5,7.4,8.8,9.5,10.5,11.8,13.2,14.5,15.8,17.2]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1700,120,42000]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

▶ from sklearn.linear_model import LinearRegression
X = [
    [1000,70,10000],
    [1200,82,15000],
    [1400,95,22000],
    [1500,105,30000],
    [1600,115,38000],
    [1800,130,45000],
    [2000,145,52000],
    [2200,160,60000],
    [2400,175,68000],
    [2600,190,76000]
]
y = [6.5,7.4,8.8,9.5,10.5,11.8,13.2,14.5,15.8,17.2]
model = LinearRegression()
model.fit(X, y)
new_sample = [[1700,120,42000]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

```

... Predicted Car Price: 10.932530174340538 Lakhs

```

DATA SET 4:

```

from sklearn.linear_model import LinearRegression
X = [

```

```

[1500,1,12000],
[1800,2,20000],
[2000,3,28000],
[2200,4,36000],
[2400,5,44000],
[2600,6,52000],
[2800,7,60000],
[3000,8,68000],
[3200,9,76000],
[3400,10,84000]
]
y = [15.5,14.8,13.9,13.0,12.0,11.0,10.2,9.4,8.7,8.0]
model = LinearRegression()
model.fit(X, y)
new_sample = [[2100,4,34000]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

from sklearn.linear_model import LinearRegression
X = [
    [1500,1,12000],
    [1800,2,20000],
    [2000,3,28000],
    [2200,4,36000],
    [2400,5,44000],
    [2600,6,52000],
    [2800,7,60000],
    [3000,8,68000],
    [3200,9,76000],
    [3400,10,84000]
]
y = [15.5,14.8,13.9,13.0,12.0,11.0,10.2,9.4,8.7,8.0]
model = LinearRegression()
model.fit(X, y)
new_sample = [[2100,4,34000]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

```

... Predicted Car Price: 13.145277774101565 Lakhs

```

DATA SET 5:

```

from sklearn.linear_model import LinearRegression
X = [

```

```

[30,250,8],
[35,300,8],
[40,350,7],
[45,400,7],
[50,450,6],
[55,500,6],
[60,550,5],
[65,600,5],
[70,650,4],
[75,700,4]
]
y = [10.5,12.0,14.0,16.5,18.5,21.0,24.0,27.5,31.0,35.0]
model = LinearRegression()
model.fit(X, y)
new_sample = [[52,470,6]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

OUTPUT:

```

▶ from sklearn.linear_model import LinearRegression
X = [
    [30,250,8],
    [35,300,8],
    [40,350,7],
    [45,400,7],
    [50,450,6],
    [55,500,6],
    [60,550,5],
    [65,600,5],
    [70,650,4],
    [75,700,4]
]
y = [10.5,12.0,14.0,16.5,18.5,21.0,24.0,27.5,31.0,35.0]
model = LinearRegression()
model.fit(X, y)
new_sample = [[52,470,6]]
prediction = model.predict(new_sample)
print("Predicted Car Price:", prediction[0], "Lakhs")

```

```

... Predicted Car Price: 20.720000000000002 Lakhs

```

RESULT

The Car Price Prediction Model was successfully implemented using Python and the Linear Regression algorithm. The model predicts the selling price of a car based on its age, mileage, engine capacity, and horsepower. The Mean Squared Error (MSE) and R^2 score are calculated to evaluate the model, and the estimated price of a new car is displayed.